

Ex. No: 10 Paging Technique

C Program:

```
#include <stdio.h>

int main()
{
    int pageTable[20];
    int pageSize;
    int numPages;
    int logicalAddress;
    int pageNumber;
    int offset;
    int frameNumber;
    int physicalAddress;
    int i;

    printf("Enter Page Size: ");
    scanf("%d", &pageSize);

    printf("Enter Number of Pages: ");
    scanf("%d", &numPages);

    printf("Enter Frame Numbers for Each Page:\n");

    for (i = 0; i < numPages; i++)
    {
        printf("Page %d -> Frame: ", i);
        scanf("%d", &pageTable[i]);
    }

    printf("Enter Logical Address: ");
```

```
scanf("%d", &logicalAddress);

pageNumber = logicalAddress / pageSize;
offset = logicalAddress % pageSize;

if (pageNumber >= numPages)
{
    printf("Invalid Logical Address\n");
    return 0;
}

frameNumber = pageTable[pageNumber];

physicalAddress = (frameNumber * pageSize) + offset;

printf("\nPage Number : %d", pageNumber);
printf("\nOffset : %d", offset);
printf("\nFrame Number : %d", frameNumber);
printf("\nPhysical Address : %d\n", physicalAddress);

return 0;
}
```

Output:

```

(kali㉿kali)-[~]
$ gcc paging.c -o paging
(kali㉿kali)-[~]
$ ./paging
Enter Page Size: 100
Enter Number of Pages: 4
Enter Frame Numbers for Each Page:
Page 0 → Frame: 5
Page 1 → Frame: 2
Page 2 → Frame: 8
Page 3 → Frame: 1
Enter Logical Address: 250

Page Number : 2
Offset : 50
Frame Number : 8
Physical Address : 850

```

Shell Script :

```
#!/bin/bash
```

```
echo "Enter Page Size:"
```

```
read pageSize
```

```
echo "Enter Number of Pages:"
```

```
read numPages
```

```
declare -a pageTable
```

```
for ((i=0; i<numPages; i++))
```

```
do
```

```
    echo "Enter Frame Number for Page $i:"
    read pageTable[$i]
done

echo "Enter Logical Address:"
read logicalAddress

pageNumber=$((logicalAddress / pageSize))
offset=$((logicalAddress % pageSize))

if [ $pageNumber -ge $numPages ]
then
    echo "Invalid Logical Address"
    exit
fi

frameNumber=${pageTable[$pageNumber]}

physicalAddress=$((frameNumber * pageSize + offset))

echo "Page Number : $pageNumber"
echo "Offset : $offset"
echo "Frame Number : $frameNumber"
echo "Physical Address : $physicalAddress"
Output:
```

```

(kali㉿kali)-[~]
$ chmod +x paging.sh
./paging.sh
Enter Page Size:
100
Enter Number of Pages:
4
Enter Frame Number for Page 0:
5
Enter Frame Number for Page 1:
2
Enter Frame Number for Page 2:
8
Enter Frame Number for Page 3:
1
Enter Logical Address:
250
Page Number : 2
Offset : 50
Frame Number : 8
Physical Address : 850

Enter deadlock process numbers (if
No Deadlock Detected

--kali㉿kali--
$ ./deadlock.sh
Enter number of processes:
6
Enter deadlock process numbers (if a
1 5
Deadlock processes:
1 5

--kali㉿kali--
$ ./bank_greater.sh
--kali㉿kali--
$ ./bank_greater.sh

--kali㉿kali--
$ ./bank_greater.sh

--kali㉿kali--
$ ./bank_greater.sh
Enter three numbers:
1 4 2
1 is the greatest

```